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Graphene Oxide/1-butyl-3-Methylimidazolium Tetrafluoroborate Nanocomposite: Synthesis, Characterization and its Application for the Electrochemical Determination of an Antihistamine and Anticholinergic Embramine

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This paper discusses the synthesis and application of graphene oxide (GO)/1-butyl-3-methylimidazolium tetrafluoroborate [BMIM][BF_4] nanocomposite. The nanocomposite, GO/[BMIM][BF_4],was characterized using XRD, XPS, EDX, FESEM, TEM, FT-IR, and Raman techniques. The dispersion of [BMIM][BF_4] ionic liquid (IL)in the graphene oxide nanochannels provided remarkable interfacial property to the glassy carbon electrode (GCE) surface. GO/[BMIM][BF_4]modified GCE sensor enhanced the anodic peak current intensity of Embramine (EMB), which was found proportional to the drug concentration within the range of 4.9 to 24.7 ng l^{-1}, with a detection limit of 1.5 ng l^{-1} and the quantification limit of 4.6 ng l^{-1}. The redox behavior at varying scan rates revealed that the electro-oxidation process of EMB at GO/IL/GCE was an irreversible diffusion-controlled process. The applicability of the proposed method was further studied for the successful quantification of EMB in the pharmaceutical formulation and human blood plasma.

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The medication involved in treating allergic conditions belongs to the class of antihistamines used widely to reduce and block chemicals released by the body, i.e. histamines. Embramine (EMB, Scheme 1) is an anticholinergic and antihistamine drug that helps cure severe allergic reactions. It is chemically designated as 2-[1-(4-bromophenyl)−1-phenylethoxy]-N,N-dimethylethanaminemebro-phenhydramine. It works by restricting the chemical messengers responsible for causing itching, inflammations, hives and other allergies.

This drug has been quantified using HPLC,[1] UV-spectrophotometry,[2,3] Capillary isotachophoresis,[4] potentiometry,[5] and visible spectrophotometry.[6]On the other hand, electroanalytical techniques can analyze small volume of drugs with trace detection ability, high precision, average handling time, lessened complexity of pre-concentration steps, involvement of non-toxic solvents, etc. These techniques can be considered superior to the aforementioned analytical techniques due to their high sensitivity, low operating cost, selectivity and ability to unfold the mechanistic aspects of the real sample analysis.

Many synthetic drugs developing today require an improved methodology and prompt detection techniques with ultra-trace detection probes. This demands the fabrication of sensors that magnify the response, enhance the electron transfer rate, and reduce the overpotential of the analyte system.

Ionic liquids (IL) act as an excellent solvent, and an electrolyte is referred to as a "designer liquid" with beneficial physicochemical properties. They provide an excellent solvent media due to their polar nature.[7]They have numerous applications in electro-organic synthesis, electrochemical devices, Oxidative polymerization, etc.[8] IL-based GO composites can provide promising opportunities due to their specific properties in many areas of research like material science, chemical science and analytical chemistry.[9] Three approaches are used for the synthesis of IL-based graphene composites. The first approach involves the graphene and ILs mixing. In the second approach, pre- rGO facilitates the polymerization of polymeric IL in situ. The third approach involves the electrochemical reduction of GO in the presence of IL. The properties of ILs can further be altered by reaction with different functional materials due to surface changes. The reduced GO have been reported to show poor electrochemical properties with IL.[10] [BMIM][BF_4] IL is an efficient and recyclable reaction medium for the synthesis of nanoparticles.[11] The strong network formed at the GO/IL interface due to dispersion of IL in GO nanochannels is responsible for the effective electrocatalytic property of the modified sensor towards EMB. Here, the hydroxyl groups present in GO play a vital role in IL flow in the nanochannels.[12] The novelty of the GO/IL based sensor has been observed in the health care monitoring system where the nanocomposite has been found to explore and quantify dopamine in the real samples.[13]

On the other hand, graphene oxide (GO) is graphene with oxygen functionalities. It acts as a suitable adsorbent owing to its high surface area, extraction capacity and good mechanical strength. The delocalized nature of the electron system in graphene provides it with excellent adsorbing ability in extracting non-polar substances.[14] The adsorbents of magnetic nature have been synthesized using GO as composites for magnetic solid-phase extraction. The process has made the analytical determination of compounds of organic and inorganic nature easier in different samples.[15–17]

Thus, this paper provides deep insights into GO/[BMIM][BF_4]/GCE sensor's fabrication and its role in the electrochemical quantification of an anti-allergic drug EMB. The structural and morphological analysis of the nanocomposite, which was produced by the ultra-sonication approach, was characterized using various analytical and spectroscopic techniques. Square wave voltammetry (SWV) showed a significant percentage of enhancements in the anodic peak current and electrochemical oxidation, using GO/[BMIM][BF_4]/GCE as compared to bare GCE. The synergism of GO and [BMIM][BF_4] IL, thus, provides a novel procedure for the determination of drugs in the pharmaceutical industry.

Experimental

Instrumentation.—Workstation Autolab PGSTAT 302 N Potentiostat controlled by Nova 1.11 electrochemical software from Eco-Chemie (The Netherlands) has been used throughout the research work, which is equipped with a one-compartment cell, comprising a three-electrode setup and a stand (Metrohm 663 VA). The three-electrode assembly in this study contained a GO/[BMIM][BF_4]/GCE and GCE as the working electrode, an Ag/AgCl (3 M KCl) as a reference electrode and Pt wire as a counter electrode. A nitrogen gas bubbler was attached for the purging in the electrochemical cell. The Frequency Response Analyzer (FRA) module was utilized for the Impedance measurements.

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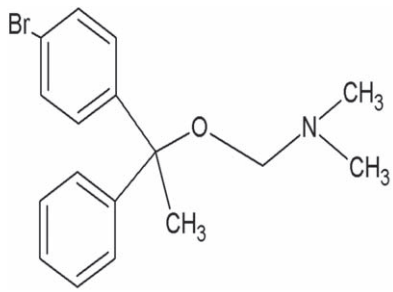


Scheme 1. Chemical Structure of EMB.

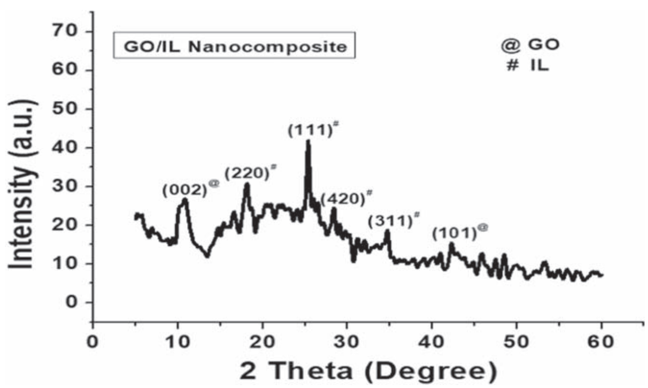


Figure 1. XRD pattern of GO and GO/IL nanocomposite.

A digital pH meter (Mettler Toledo) with an Ag/AgCl reference and a glass electrode as a working electrode was utilized to determine the pH of the solutions. X-ray diffraction (XRD) was performed at Bruker, D-8 Advance, 2008. FTIR and Raman spectra were procured on Cary 630 FTIR Spectrometer and Laser Confocal Raman Spectrometer, respectively. Field emission scanning electron microscopy (FESEM) was carried out at Carl Zeiss, 20000X, with an Energy Dispersive X-ray spectrometer (EDX).

Reagents and chemicals.—[BMIM][BF_4] ⩾ 97.0% (HPLC) has been directly purchased from Sisco Research Laboratories Pvt. Ltd. EMB standard was procured from a commercial source (India). The pharmaceutical formulation of EMB (Mebryl 25 mg) in tablet form was obtained from a commercial source. N, N-Dimethyl formamide (DMF) ⩾ 99.0% (HPLC) was purchased from Sigma Aldrich. Ultra-pure Milli-Q water used with the resistivity of 18 MΩ.cm obtained from ELGA purification system (U.K). All the other chemicals used were of analytical grade and were used without further purification.

Synthesis of GO/[BMIM][BF$_4$] nanocomposite.—GO was synthesized using well known Hummer's method. Further, the sonication method was employed to prepare GO/[BMIM][BF_4] nanocomposite. The amount of GO and IL was varied in different percentages, as shown in Fig. 9. Nanocomposites were prepared in five compositions named as A, B, C, D, E and F which were described as 1 mg GO + 100 μl IL, 0.5 mg GO + 100 μl IL, 1.5 mg GO + 150 μl IL, 1.5 mg GO + 100 μl IL, 1.5 mg GO + 50 μl IL and 2 mg GO + 100 μl IL respectively. DMF was used as the solvent medium for preparing the suspension, which was then sonicated for 120 min, followed by vortexing before use.[18]

Fabrication of GO/[BMIM][BF$_4$]/GCE.—The GCE surface was cleaned using alumina slurry (particle size 0.3 μm), spread over the Buehler cloth. The electrode was washed until a shiny appearance of GCE was observed. A 7.5 μl volume load of optimized D composition (1.5 mg GO + 100 μl IL) of the modifier was withdrawn and dropcoated over the working surface of GCE to get GO/IL/GCE. Even after several flushing with ultra-pure deionized water, the modifier was not removed from the surface of GCE.

Working analytical procedure.—The stock solution of EMB (2 mg ml^{-1}) was prepared in DMF. Working solutions were prepared by further dilution with the supporting electrolyte to get the desired concentration range. 0.2 M Britton-Robinson (B-R) buffer (pH 7.9) was used as a supporting electrolyte. About 10 ml of the electrolyte solution containing an appropriate amount EMB sample was added to the electrolytic cell in which the electrodes were immersed. N_2 gas was purged in the electrochemical cell to remove the dissolved oxygen. The voltammetric results were scanned with a

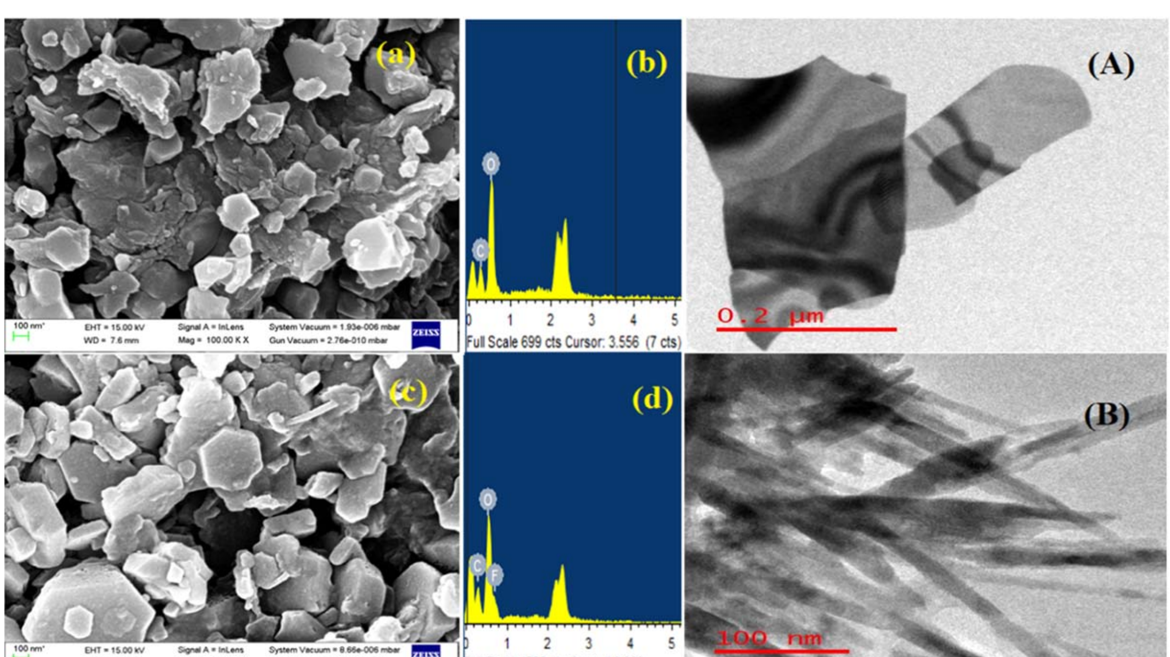


Figure 2. FESEM, EDX and TEM images of GO (a), (b), (A) and GO/IL (b), (c), (B).

potential window ranging from 0 to 1.2 V. All the experiments were performed at room temperature ±2 °C.

Pharmaceutical formulation and human blood sample assay procedure.—One tablet was weighed and finely powdered using a mortar and pestle. An adequate amount of this material equivalent to 50 mg of EMB was transferred in 25 ml flask containing ethanol. The solution was mixed thoroughly and sonicated for 30 min. Then, it was centrifuged at 500 rpm to obtain a clear supernatant containing the active ingredient. An aliquot of this sample was diluted with supporting electrolyte in the electrochemical cell and subjected to voltammetric analysis. Similarly, 1 ml of fresh human blood serum samples of a healthy volunteer was collected and was diluted almost 50 times with buffer to avoid matrix interference. This solution prepared was then analyzed, without any further treatment.

Results and Discussion

Characterization of GO/IL nanocomposite.— *Morphological and structural analysis.*—The XRD was used for characterizing the sample. The sharp peaks confirmed the presence of GO in the sample at 2θ of 10.77°, 42.68° due to [002] and [101] planes depicted by symbol @ in Fig. 1. Some other peaks at 2θ of 17.85°, 25.54°, 28.46° and 34.69° correspond to (220), (111), (420) and (311) lattice planes, which might be due to nanocomposite formation represented by symbol # in XRD spectra.[19]

The FESEM micrographs of synthesized GO and GO/IL is shown in Figs. 2a–2c.[20,21] Few hexagonal-shaped structures are observed in the GO/IL nanocomposite. This suggests the dispersion of GO in IL with improved crystallinity of the nanocomposite. Further, EDX results show the presence of C, O in Fig. 2b and the presence of C, O, F, in Fig. 2d in the GO/IL matrix. These graphs portray the presence of IL and elements occurrence during the reagents used to synthesize GO.

TEM images signify nanochannels structure of the composite where IL interact with GO, and it can be observed in Fig. 2A that graphene oxide has a layered structure, which affords ultrathin and homogeneous graphene films. Such films are folded or continuous at times and it is possible to distinguish the edges of individual sheets, including kinked and wrinkled areas. Figure 2B clearly shows the possibility of the dispersion phenomenon of the IL in the GO nanochannels.[22,23]

Chemical stoichiometry.—XPS is used to investigate the chemical composition and nature of the compound under study. GO and its nanocomposite with IL, i.e. [BMIM][BF_4] was characterized for their compositional nature by XPS (Fig. 3). The specific peaks observed are C1s, O1s, B1s and N1s, indicating their main

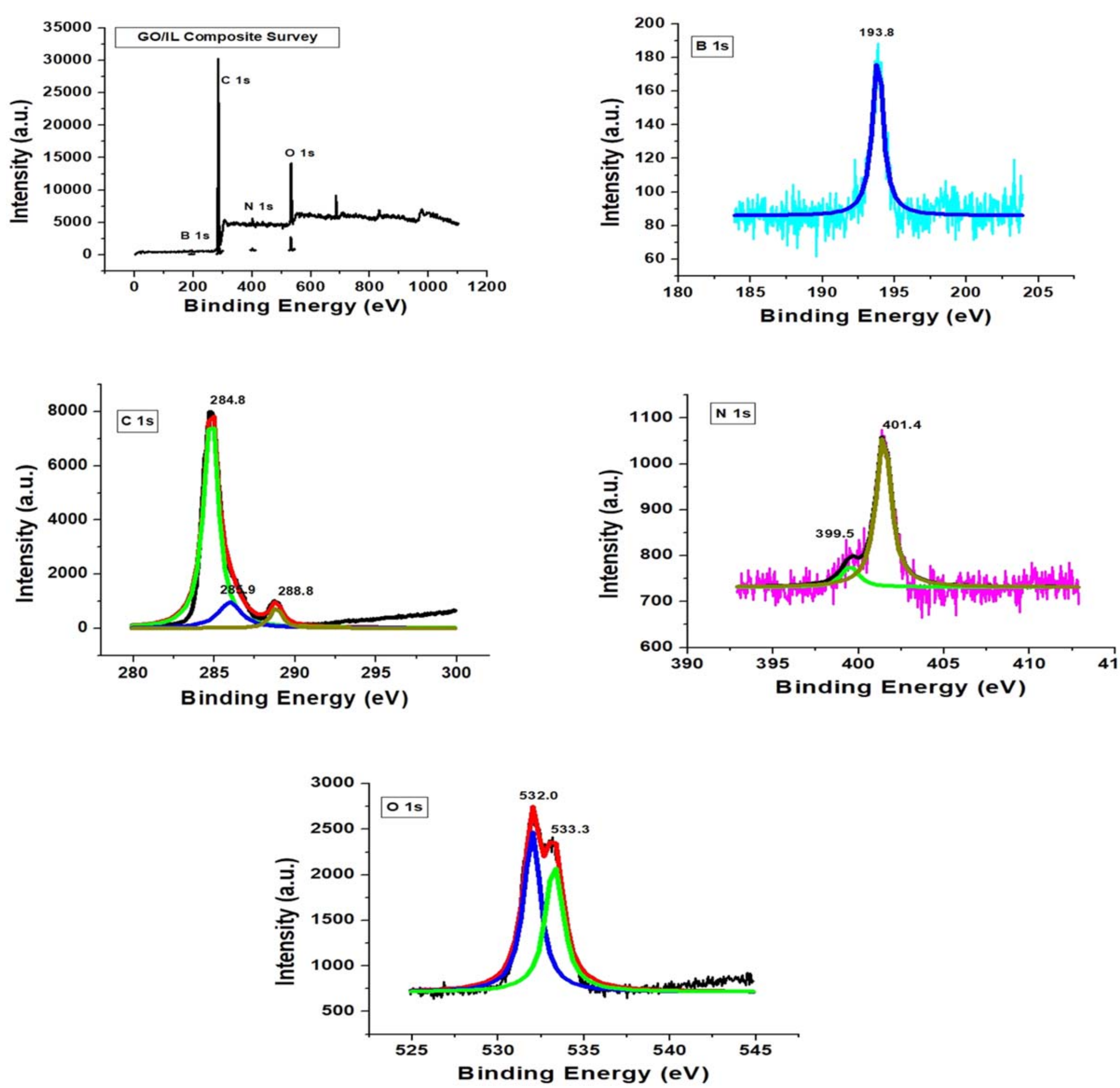


Figure 3. Full range XPS spectra and High-resolution C 1s, B 1s, N 1s and O 1s XPS spectra for GO/IL nanocomposite.

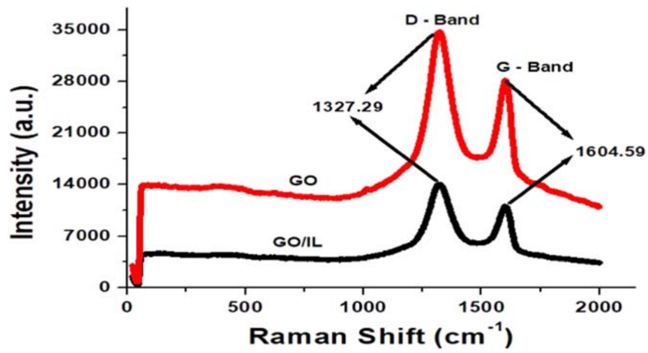


Figure 4. Raman spectra of GO and GO/IL nanocomposite.

constituents. Three characteristic peaks were observed in the C1s spectra at 284.8, 285.9 and 288.8 eV, which can be attributed due to the C–C bond in the non-oxygenated ring, C–OH bond and O=C–OH group, respectively.[24] The N1s spectra have shown two specific peaks at 399.5 and 401.4 eV, which can be assigned to quarternary ammonium group and neutral nitrogen in pyrrole ring and might be due to the presence of IL in the composite.[25] A characteristic peak of B1s can be observed in [BMIM][BF_4] at 193.83 eV. The O1s spectra have shown two peaks at 532.0 and 533.3 eV due to surface adsorbed oxygen.[26]

Functional group analysis.—The carbon-based materials are widely characterized using Raman spectra since they are non-destructive. The electronic properties and structural information on materials can be easily obtained. Figure 4 shows a typical Raman spectrum for GO and GO/IL nanocomposite. The two bands (or peaks), i.e., D- and G- band, have been positioned at 1327.29 cm^{-1} and 1604.59 cm^{-1}, respectively, in the Raman spectrum of GO. These two characteristic peaks are the main features in the Raman spectrum of GO.[27] The two bands indicate vibrational properties of the material and arise due to the sp^2 nature of covalent bonds. The D-peak arises due to breathing-like modes while G-peak corresponds to E_{2g} phonons (optical modes) which are Raman active.[28]The decrease in peak intensity may be attributed to [BMIM][BF_4] interaction with GO. The peak intensity of the D-band also reflects the amount of disorder.[29]

FTIR gives information about the chemical and structural nature of the material under study. The characteristics of peak positions indicate a specific functional group. FTIR spectrum (Fig. 5a) was recorded for GO and GO/IL nanocomposite in the spectrum range of 500 to 4500 cm^{-1}. In the FTIR spectrum of GO, the peaks observed at 1115.50, 1396.38, 1593.29 cm^{-1} indicate alkoxy C–O, epoxy C–O and aromatic C=C bonds, respectively. A slight broad peak above 3000 cm^{-1} depicts the absorption of water molecules in GO layers. The characteristic peak observed at 3430.47 cm^{-1}represents the stretching of hydroxyl O-H bonds.[30]

The FTIR spectrum (Fig. 5b) of GO/IL has shown characteristic peaks. The peak observed at 2886.71 cm^{-1}was assigned to aliphatic symmetric stretching vibrations of the C-H group in [BMIM][BF_4]. The peaks at 1237.45 and 1096.51 cm^{-1} correspond to methyl groups' in-plane bending vibrations. The peaks observed at 1621.28 and 1424.36 cm^{-1} depict C=C and C=N stretching vibrations. The peak at 609.72 cm^{-1}is attributed to C–N stretching vibration.[31]

Electrochemical characterization of the sensors.— *Electroanalytical behaviour of EMB at GO/IL/GCE.*—The SWV

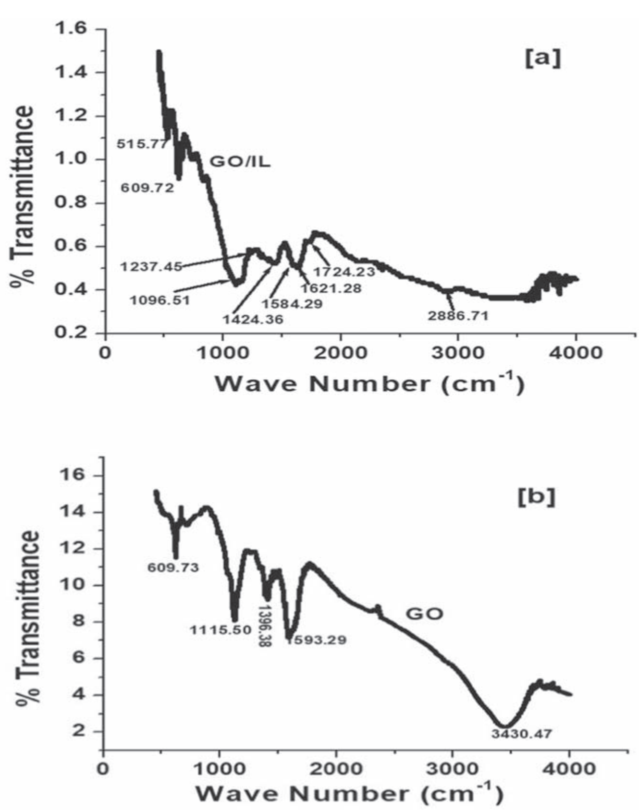


Figure 5. FTIR spectra of GO and GO/IL nanocomposite.

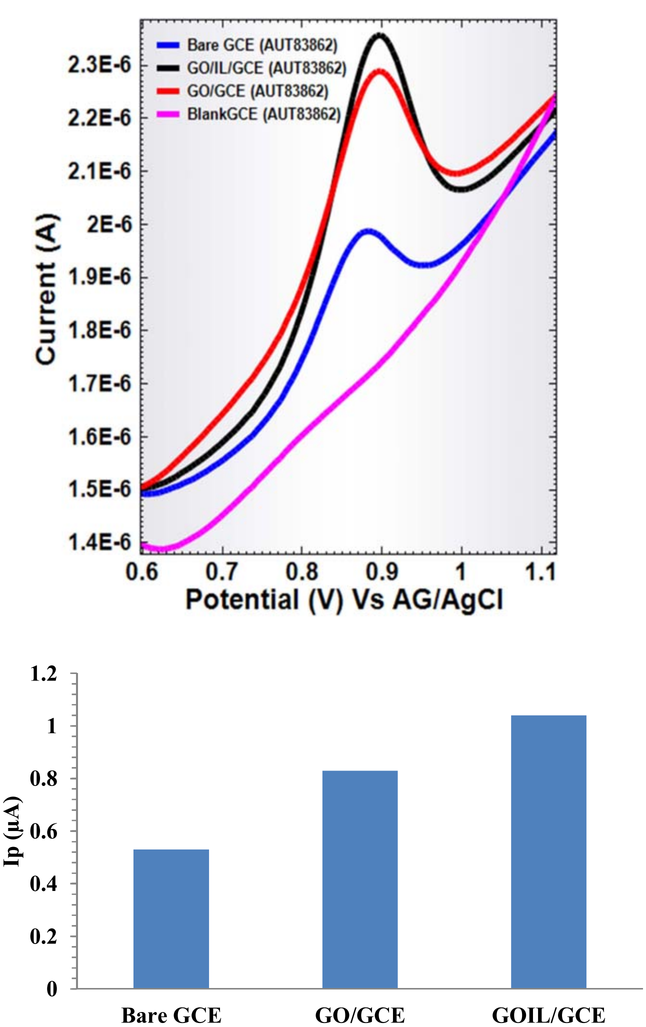


Figure 6. SWV for determination of EMB at pH 7.9, B-R buffer at bare and modified GCE. Bar Graph showing peak current values using different modifiers and GCE.

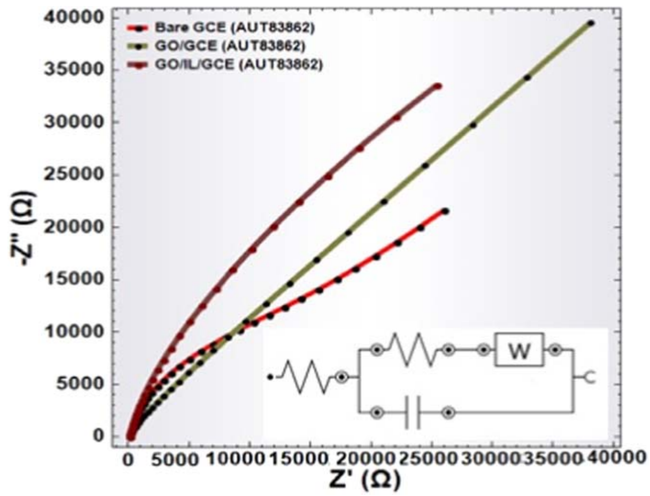


Figure 7. Nyquist plots of 5 mM $Fe(CN)_6^{3-/4-}$ in 0.1 M KCl at GCE, GO/GCE, and GO/IL/GCE and corresponding inset equivalent circuit used for the determination of charge transfer resistance.

technique investigated the voltammetric behaviour of EMB at GCE and the modified electrode. Figure 6 shows the SWV of EMB with enhanced anodic peak current intensity at GO/IL/GCE. The response characteristics of SWV exhibits that the anodic peak current intensity was increased by almost 56% and 96% at GO/GCE and GO/IL/GCE, respectively, compared to the bare GCE. This can be attributed to the dispersion of the [BMIM][BF_4] in GO channels leading to the high surface area, which results in the trapping of the solution containing EMB and enhanced peak current intensity.[32]

Nanocomposite GO/IL/GCE was optimized and applied for the rest of the measurements based on prominent peak shape and enhanced peak current intensity.

Electrochemical Impedance Spectroscopy (EIS).—EIS technique was used to investigate the electronic transfer phenomenon occurring at GO/IL/GCE in a $K_3[Fe(CN)_6]$ solution. Nyquist plots obtained at different modified electrodes are shown in Fig. 7. The semi-circle region signifies the electron transfer limited process and hence the charge transfer resistance, whereas the linear portion represents the diffusion process. The circuit (Inset) comprises Rs the solution resistance, C the double layer capacitance, R_{ct} the charge transfer resistance and W the Warburg impedance. The Warburg type impedance represents the linear line that appeared at lower frequencies because of the diffusion-controlled process. The R_{ct} value for bare GCE was found to be 14.3 kΩ, 5.6 kΩ for GO/GCE and 1.8 kΩ for GO/IL/GCE, respectively. A decrease in the charge transfer resistance and increased conductivity is observed in the case of GO/IL/GCE compared to the bare GCE.[33]

Electroactive surface area.—The electrocatalytic activity of the GO/IL/GCE was scrutinized using the CV technique. Figure 8 shows Cyclic Voltammogram recorded for potassium ferricyanide solution (1 mM in 0.1 M KCl) at the bare GCE, GO/GCE and GO/IL/GCE. Enhanced current values at GO/IL/GCE compared to bare GCE are observed with well-defined redox peaks of the analyte system. The surface areas were then determined using the Randles-Sevcik equation as follows:

$$I_p = (2.69 \times 10^5)n^{3/2}AC_oD^{1/2}\nu^{1/2} \quad [1]$$

Where I_p refers to the anodic peak current (ampere), D the diffusion coefficient ($cm^2\ s^{-1}$), A the surface area (cm^2), ν the scan rate ($V\ s^{-1}$), C_o the concentration of the redox probe ($M\ cm^{-3}$) and n the number of electron transfer involved. For 1 mM $K_3Fe(CN)_6$, $n = 1$ and the diffusion coefficient $D = 7.60 \times 10^{-6}\ cm^2\ s^{-1}$. On substituting values in the above equation, the surface area of

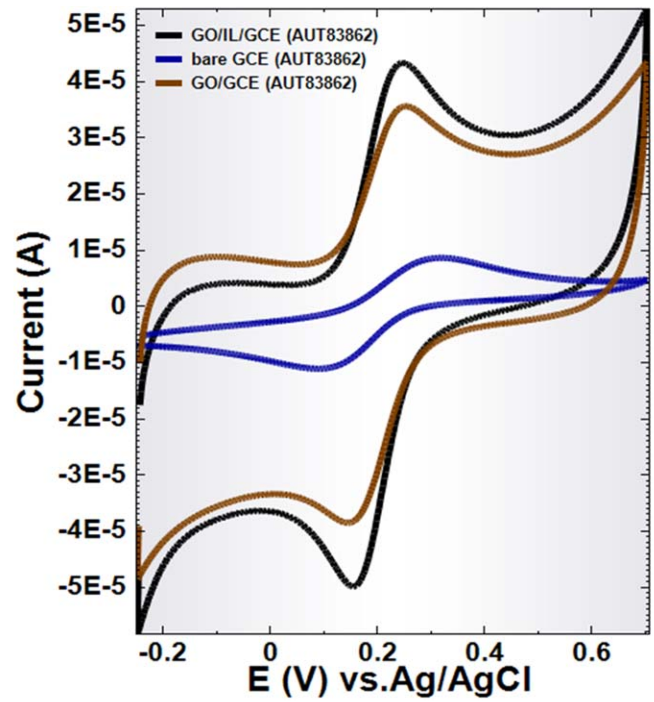


Figure 8. CV of 1 mM $K_3[Fe(CN_6)]$ in 0.1 M KCl at bare, GO/GCE and GO/IL/GCE (scan rate 100 mV s^{-1}).

GO/IL/GCE, GO/GCE, and bare GCE were found to be 0.21 cm^2, 0.10 $cm^2$and 0.042 cm^2, respectively. Thus, after the surface modification, the electroactive surface area of the electrodes was considerably increased.[34]

Effect of modifier amount.**—**Six compositions, namely A, B, C, D, E and F, were prepared as described in the experimental section (Fig. 9a). Out of these five prepared composites, "D" showed the highest peak current values, which signify better sensitivity as a sensor.[35]

For GCE modification, "D" composite with the optimized volume of the nanocomposites was utilized for further electrochemical studies. The load volume used with D composite was 2 μl, 5.0 μl, 7.5 μl, 10.0 μl,12.5 μl and 15 μl. Optimized results showed a 7.5 μl load, which resulted in the highest peak current values and prominent shape, as shown in Fig. 9b.

Effect of supporting electrolyte and pH.**—**Analyte redox behaviour is influenced by nature and the pH of the supporting electrolyte. Using different buffer solutions, the CV study of EMB at GO/IL/GCE showed that B-R buffer gave the best response. Hence, electrocatalytic oxidation of EMB in 0.2 M B-R buffer as an electrolyte was studied within the pH range of 2.5–12. The enhancement in anodic peak current intensity was optimized at 7.9 pH. Therefore, the B-R buffer at pH 7.9 was selected as the optimized pH in all voltammetric measurements.

Further, the anodic peak current of EMB was sensitive to the pH of the solution. The shift in the anodic peak potential values with the augmentation of solution pH was observed (Fig. 10), indicating the role of participated protons in the electrode reaction process of EMB. A linear relationship of peak potential on pH of the supporting electrolyte was obtained following the equation:

$$\mathrm{Ep(V)(vs\ Ag/AgCl)} = -0.06\ \mathrm{pH} + 1.38;\ R^2 = 0.99 \quad [2]$$

Slope of 61 mV pH^{-1} in the plot of Ep(V) vs pH explains that the value is close to the expected value of 59 mV pH^{-1}, which clearly indicates that the proposed number of protons and electrons participating in the electrode process are equal.

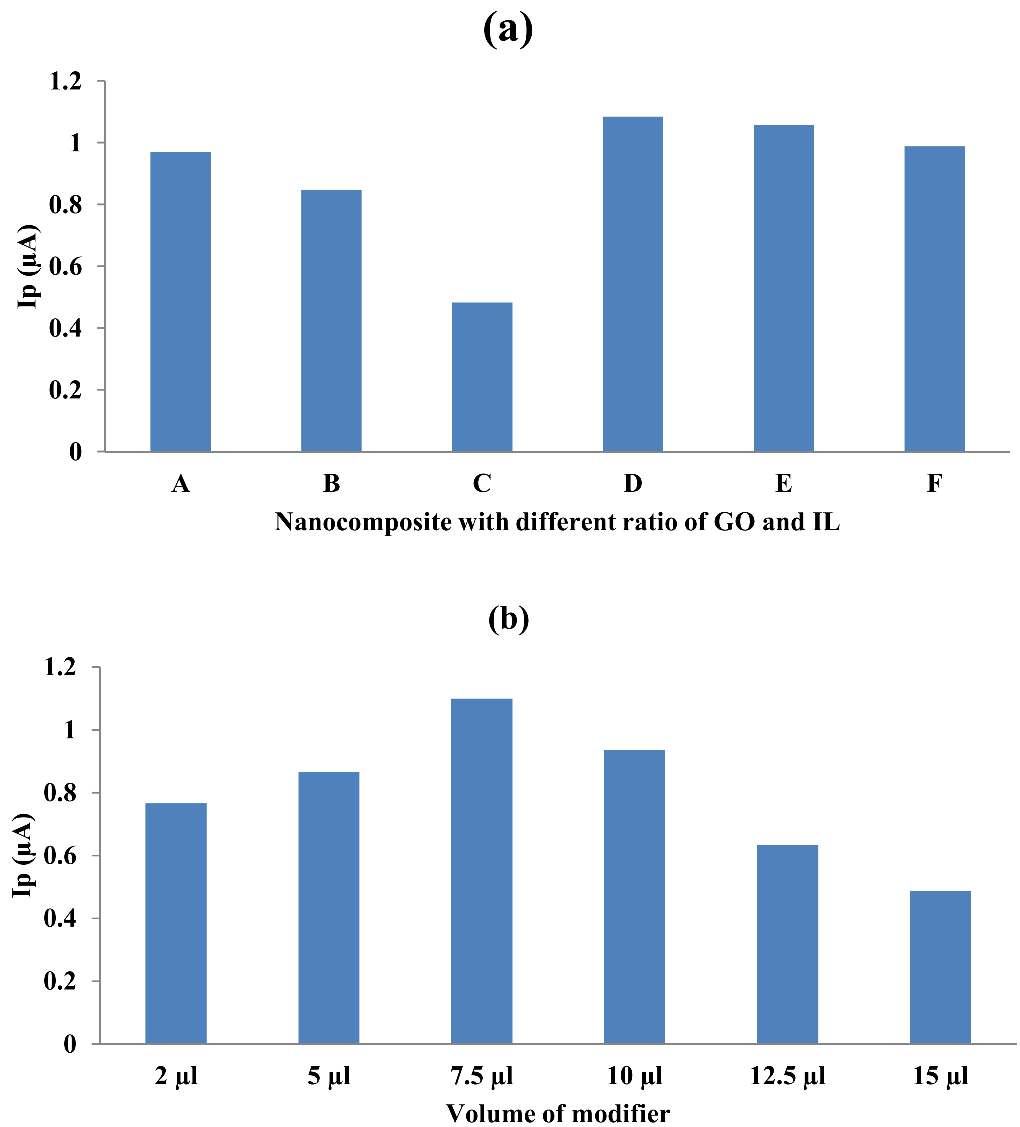


Figure 9. Bar graph showing (A) Different ratios of GO and IL used to optimize current values of the analyte system. (B) The current response of EMB (pH 7.9, B-R buffer) with different volumes of modifier (GO/IL nanocomposite) dropcoated on GCE.

Effect of scan rates.—Electrocatalytic mechanistic aspects and the redox information are well elucidated by the response characteristics of CV at different scan rates. Figure 11 shows the influence of varying scan rates (10–200 mV s^{-1}) on the anodic electro-oxidation process of EMB (5.90 ng l^{-1}) at GO/IL/GCE. A plot of peak current intensity (Ip) and the square root of scan rate ($\nu^{1/2}$) were obtained, following the linear equation:

$$\mathrm{Ip}(\mu\mathrm{A}) = 1.75\nu^{1/2}(\mathrm{V/sec}) + 0.19;\ \ \mathrm{R}^2 = 0.98 \tag{3a}$$

$$\mathrm{Ln\ Ip}(\mu\mathrm{A}) = 0.40\ \ln \nu(\mathrm{V/sec}) + 0.06;\ \ \mathrm{R}^2 = 0.97 \tag{3b}$$

Equations 3a and 3b suggest a diffusion-controlled process of EMB at the GO/IL/GCE electrode interface.

Hence, the electro-oxidation phenomenon of EMB at GO/IL/GCE was irreversible (Fig. 11 as no cathodic peak was observed. Therefore, no significant shifting of peak potential values was observed with the increase of ν (V/sec) that a linear equation can express:

$$\mathrm{Ep\ (V\ vs\ Ag/AgCl)} = 0.05\ \ln \nu(\mathrm{V/sec}) + 0.95;\ \mathrm{R}^2 = 0.97 \tag{4}$$

Thus from Eq. 4 (Fig. 11c) and according to the Laviron equation, for an irreversible anodic behaviour[36,37]

$$RT/\alpha nF = 0.051(\text{For the electrode process at 298 K}) \tag{5}$$

where, T = 298 K, R = 8.314 J K^{-1}.mol^{-1} and F = 96,500 C. The value of α.n can be calculated from the slope and was found to be 0.51. The electron transfer coefficient 'α' is assumed to be 0.5 for an irreversible diffusion-controlled process. The value of n, therefore, comes out to be ~1.

Moreover, the Nernst equation employed to find the number of protons involved in the reaction process can be observed by the following equation:[38]

$$Ep = Ep^0 - 2.303RTmpH/(1 - \alpha)nF \tag{6}$$

Here "m" is the number of protons participating in the electrode reaction, which is 1.

Proposed reaction mechanism of EMB.—The study of the voltammetric behaviour of EMB concludes that it is an irreversible diffusion-controlled anodic reaction that occurs in B-R buffer; pH 7.9, at GO/IL/GCE. This involves the loss of one electron and one proton due to N's presence in the compound as the site for oxidation leading to the formation of oxidized EMB (Scheme 2). This

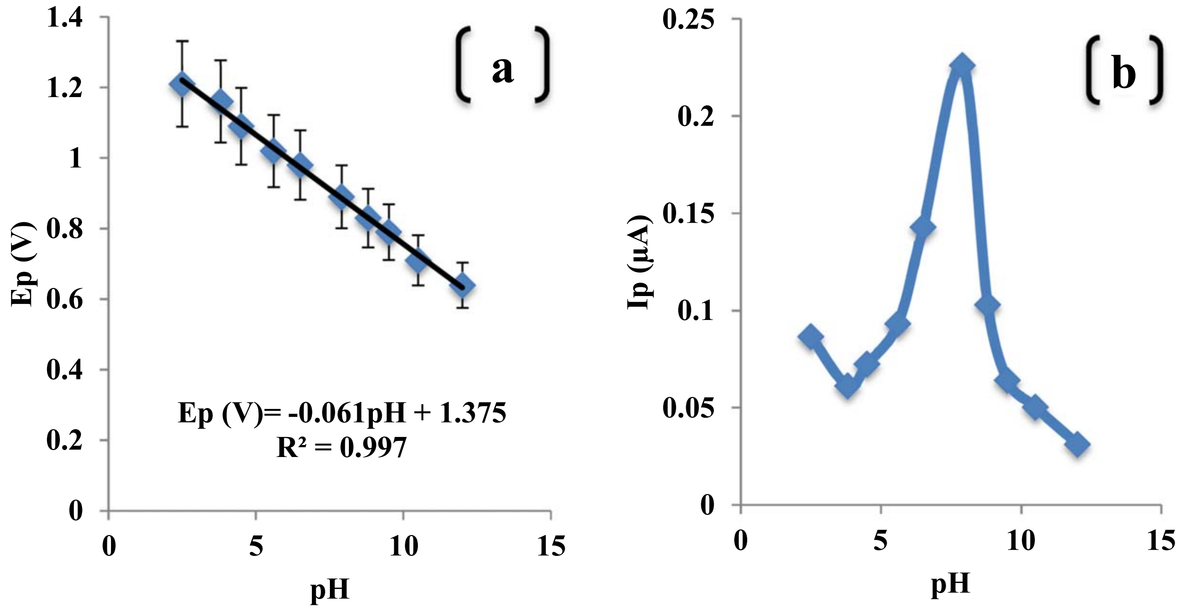


Figure 10. Linear plot of (a)oxidation peak potential Ep vs pH; (b)oxidation peak current Ip (μA) vs pH in 0.2 M B-R buffer at varying pH value (2.5–12.0) at scan rate 100 mV s^{-1}.

Table I. Reproducibility and repeatability results of EMB under optimized conditions.

	Reproducibility		Repeatability	
Sensor	Mean Ip (μA)[a]	% RSD[a]	Mean Ip (μA)[c]	% RSD[c]
1	0.37	1.17	0.3641	1.76
2	0.36	0.82		
3	0.34	1.21		
4	0.37	1.31		
5	0.32	0.93		
Avg[b]	0.35	1.08		

a) Average of 5 observations. b) Averageof five days observations. c) Average of 4 samples.

Table II. Accuracy and precision.

St. Add	St. Found[b]	%COV[a]	Precision	RE%
5	4.91	0.56	4.91 ± 0.027	−1.88
7	6.94	0.49	6.95 ± 0.034	−0.77
10	9.96	0.36	9.96 ± 0.036	−0.39

"St." belongs to standard a) Coefficient of variation = S.D./Mean x100; b) Amount found represents mean of six observations.

Table III. Recovery analysis in pharmaceutical formulation.

St. Add	St. Found	% RSD	% Recovery
5	4.99	0.408	98.97
7	6.97	0.303	98.95
10	9.98	0.151	99.38

mechanism is supported by the Sanghavi et al. 2013 experimental work,[39] which again favours the proposed mechanism of the antihistamine drug.

From the literature and based on all the experimental phenomena, a probable oxidation mechanism of EMB on GO/IL/GCE has been proposed, as shown in Scheme 2.

Method validation, calibration curve and detection limit.— Based on the observations, SWV was recorded with the increasing amounts of EMB, which under optimized conditions, the peak currents increased linearly with the increasing concentration (Fig. 12). Linearity between the peak current and the EMB concentration was observed in the range of 4.9 ng l^{-1} to 24.7 ng l^{-1}. The calibration plot of Ip (μA) vs concentration (ng ml^{-1}) is described by the equation:-

$$Ip(\mu A) = 0.028\, C_{EMB}(ng\ ml^{-1}) + 0.048;\ R^2 = 0.99 \qquad [7]$$

The detection limit (LOD = 1.5 ng l^{-1}) and quantification limit (LOQ = 4.6 ng l^{-1}) were calculated from 3 S m^{-1} and 10 S m^{-1} respectively, where "m" being the slope of the calibration curve and "S" is the standard deviation.

The sequential measurement of EMB scrutinized the reproducibility of the fabricated electrode at GO/IL/GCE by SWV (Table I). Results are satisfactorily reproducible, which can be observed from the values corresponding to the relative standard deviation of 1.08%. Moreover, to investigate the stability of GO/IL/GCE, the modified electrode was stored at room temperature for a week. The peak current response of EMB showed a retained response of 95% after seven days. These results indicated that GO/IL/GCE possess good stable and reproducible results towards the selective determination of EMB. However, almost 15% of its initial response was lost after 15 days.[40]

Effects of some common excipients like glucose, sucrose, starch were studied to explore the analytical application and specificity of the method. None of these excipients (added in excess amount) was found to interfere with the redox potential of EMB without affecting the performance of the sensor (Table V). A fixed amount of EMB was used in the electrochemical cell, and further analysis was performed by the proposed method. The obtained mean percentage recoveries based on the average of six replicate measurements were 98%. When heavy metals were checked for interference with Pb^{2+}, Cd^{2+}, Pd^{2+}, In^{2+}, Bi^{3+}, Ni^{2+}, Co^{2+} with almost 50 fold increase in concentrations, not much effect was observed regarding peak current response of the analyte system showing a variation of nearly ±2% change in values.[41]

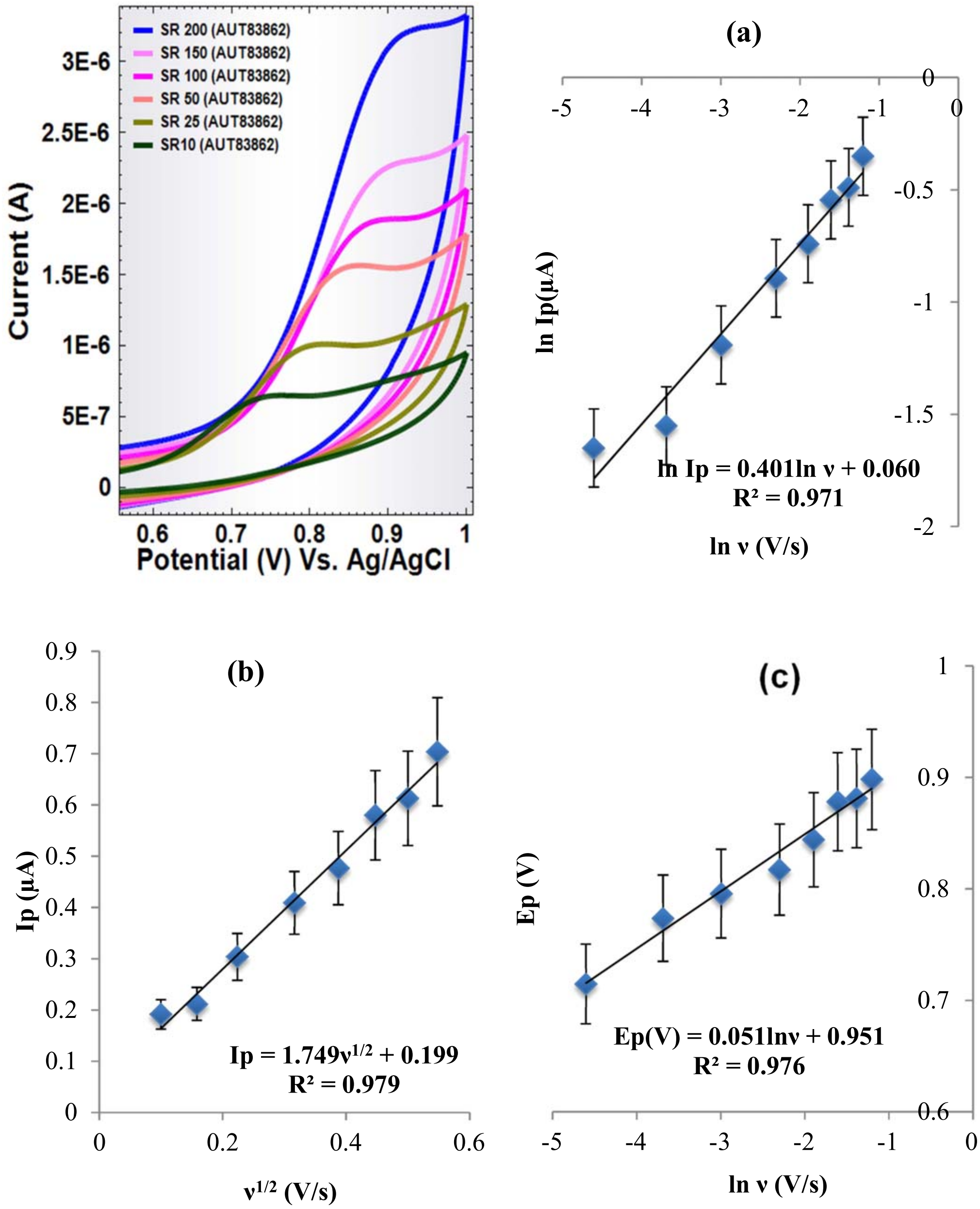
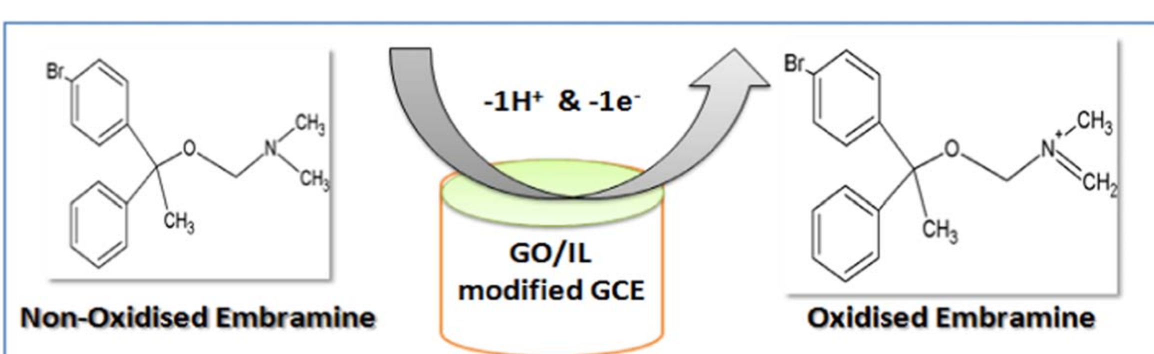


Figure 11. CV of EMB at GO/IL/GCE in B-R buffer (pH 7.9) at different scan rates (10–200 mV s^{-1}) and plot(b) Ip vs $v^{1/2}$, (c) Ep vs ln v.



Scheme 2. Probable oxidation mechanism of EMB

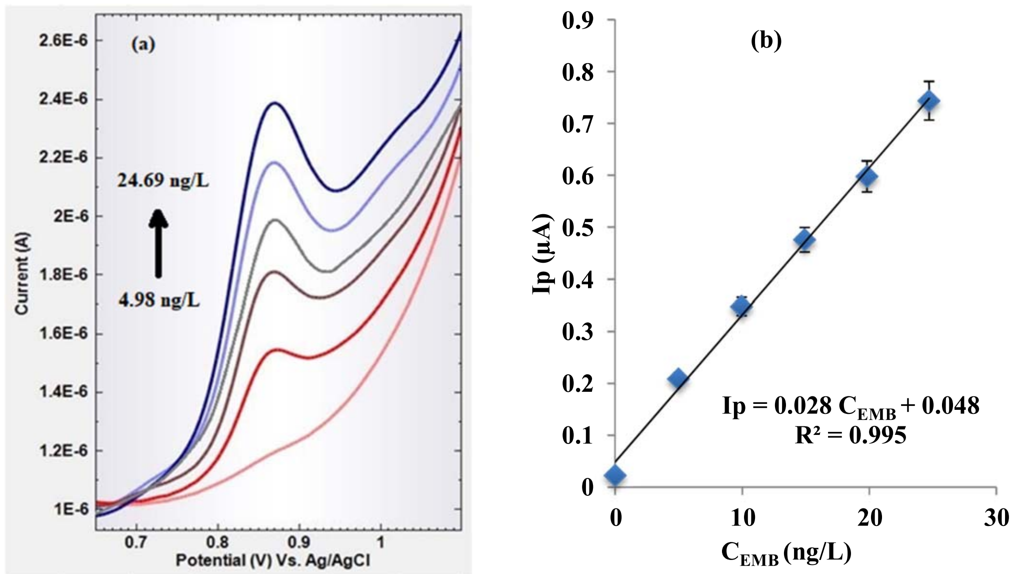


Figure 12. (a) SWV of EMB at different concentrations (4.9–24.7 ng l^{-1}) in B-R buffer at pH 7.9; b) Plot of Ip (μA) vs concentration (ng/L).

Table IV. Recovery analysis in blood serum.

St. Add	St. Found	% RSD	% Recovery
5	4.96	1.49	99.39
7	7.01	0.41	100.09
10	9.97	0.13	99.71

Table V. Excipients Interference study on the oxidation behaviour of EMB under optimized conditions.

Index	Species	Conc. (μg l^{-1})	% Change in signal
A	Glucose	1.247	−1.56
B	Fructose	1.247	−0.90
C	Sucrose	1.247	−0.35
D	Lactose	1.247	−0.27
E	Starch	1.247	−0.30
F	Acetic Acid	1.247	1.26
G	Ascorbic Acid	1.247	1.05
H	Na^+	2.494	−0.52
I	Ca^{2+}	2.494	−0.56
J	K^+	2.494	−0.25
K	CO_3^-	2.494	1.35
L	NO_3^-	2.494	0.95
M	Pb^{2+}	0.499	−0.27
N	Cd^{2+}	0.499	−0.23
O	Pd^{2+}	0.499	−0.41
P	In^{2+}	0.499	−0.39
Q	Bi^{3+}	0.499	−0.32
R	Ni^{2+}	0.499	−0.42
S	Co^{2+}	0.499	−0.39

Thus, the data obtained from the experimental results conclude that the fabricated GO/IL/GCE can be successfully optimized for the successful assay of EMB in the pharmaceutical formulation.

Analytical applications.—*Assay of EMB in pharmaceutical dosage and human blood serum form.*—Applicability of the proposed method under optimized electrode dynamic parameters was used for determining EMB present in the tablet dosage (Mebryl, 25 mg) using standard addition method. The solution system of the formulation was prepared according to the procedure mentioned in the previous section. Further, the content was finally analyzed using GO/IL/GCE by SWV and recovery analysis was performed. The average recovery of EMB (Table III) was found to be 99% in pharmaceutical formulation, which showed that the determined amount of EMB was in agreement with the claimed amount. Furthermore, the recovery results (Table IV) in human blood serum analysis showed a recovery percentage ranging from 99.1% to 101%. The accuracy of the method was evaluated by the standard addition method, spiking different concentrations of EMB in the sample solution. Percentage relative error with accuracy and precision has been reported in Table II.[42]

Conclusions

Voltammetric behaviour of EMB was studied using a fabricated novel sensor, i.e. GO/IL/GCE. The GO/[BMIM][BF_4] modified GCE was successfully characterized by XRD, FESEM, EDX, XPS, FTIR, RAMAN and EIS methods. The electro-oxidation of EMB was achieved at 0.85 V, which was seen as dependent on the solution pH of the supporting electrolyte (B-R buffer). Based on all the electroanalytical calculations and measurements, a plausible oxidation mechanism of EMB has been deduced. The possible applicability of the proposed method with specificity, accuracy, higher sensitivity, reproducibility was effectively studied and was applied for the estimation of EMB in pharmaceutical samples.

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